

FCC3 246 – CONCRETE MATERIALS (45HRS)

Prerequisites

1. SPH3 107: Physics IA
2. SPH3 108: Physics IB
3. SCH3 181: Chemistry IA
4. SCH3 182: Chemistry IB

Course Purpose

This course will provide learners with knowledge on the manufacture process of cement, manufacture and properties of concrete, testing of concrete and durability properties of concrete.

Expected Learning Outcomes

At the end of the course, a learner should be able to:

1. Understand the intricacies of the process of the manufacture of cement, including quality control of the manufacture process as well as the resulting environmental concerns.
2. Understand the process of manufacture of concrete, quality control of the process, testing of concrete and properties of both plastic and hardened concrete.
3. Understand durability aspects of hardened concrete including concrete thermal properties and corrosion of steel in concrete.

Course Content

1. Manufacture and properties of inorganic cements – Portland cement
2. Hydration of Portland cement
3. Rheology of fresh concrete – workability and segregation
4. Properties of hardened concrete – concrete compressive and tensile strengths, creep and shrinkage
5. Manufacture of concrete – ready mix concrete
6. Concrete mix design
7. Non-destructive testing of concrete
8. Durability of concrete -durability quality control of concrete, corrosion of steel in concrete, thermal properties of concrete, effects of fire on concrete
9. Materials Laboratory

Mode of Delivery

1. 2 Hour lectures per week
2. 2 Hour Laboratory sessions per week throughout the semester

Laboratory Exercises

1. Preparation of Concrete C20/25 (KS EN 1992 designation) Grade 25 (BS 8110 designation) concrete
2. Testing of plastic concrete – slump and compaction factor
3. Testing of hardened concrete – split tensile and compressive strength tests

Assessment:

1. The course shall be examined as follows:
2. 2 Hour written Examination at the end of the semester (70%)
3. Continuous Assessment Tests (20%)
4. Written Laboratory Reports (10%)

Core Reading Materials

1. Neville A M (2007) *Properties of Concrete*, 5th Edition